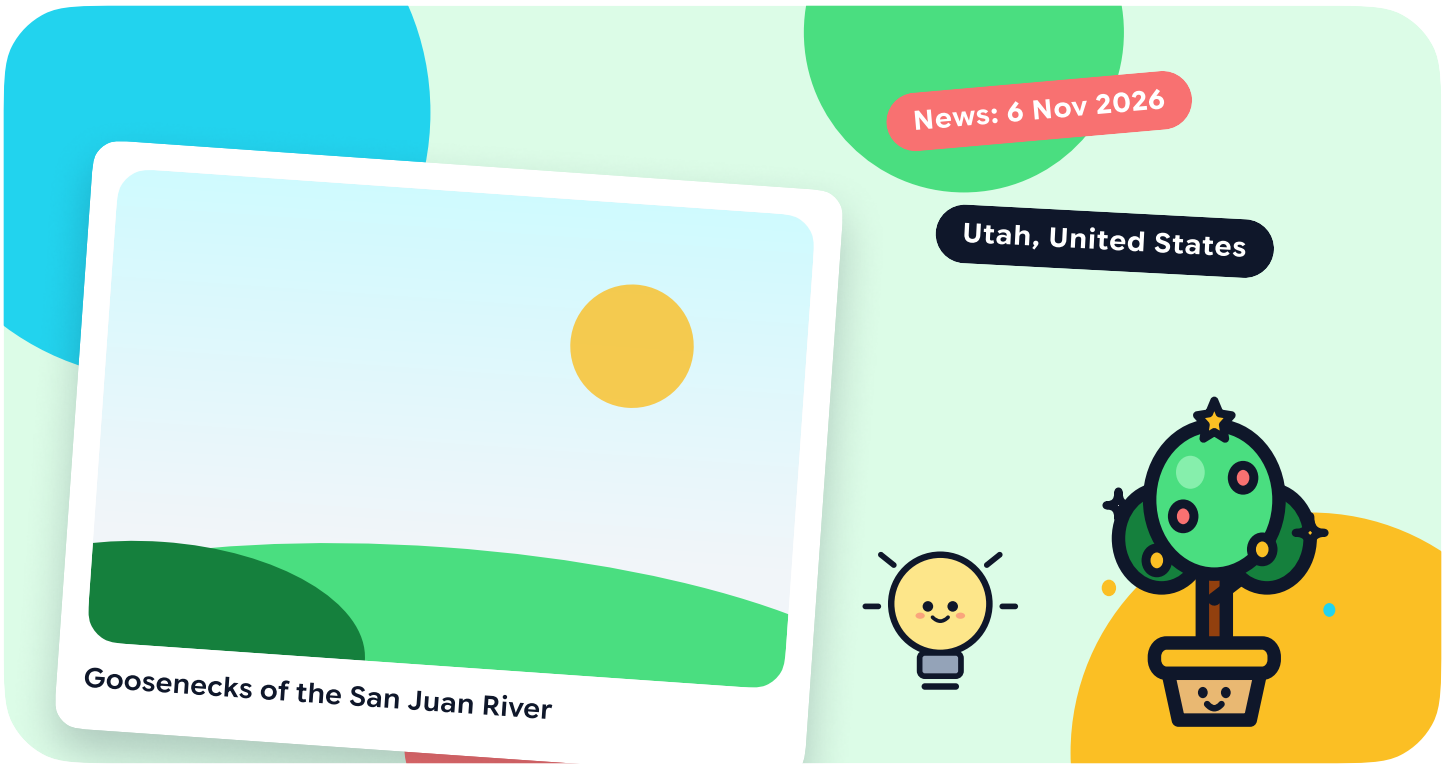
#02 · 9 NOV 2026 EN +60 XP



Potential energy · Physics · Energy

Carving stone with water

How does a winding river cut deep canyons through ancient rock? Discover the gravitational energy behind river meanders.

YOUR QUEST MAP

STOP 1 p.2
The story
Read how rivers carve canyons

STOP 2 p.3
The physics
Use Ep = m × g × h

STOP 3 p.4
Your mission
Quiz, mini lab and badge

12 min read · SMP · grade 7–9 · Earn up to 60 XP

DID YOU KNOW?

The San Juan River travels 5.5 miles between two points that are only 1.3 miles apart as the goose flies. [1]

STOP 1 · THE STORY

+10 XP FOR READING

The winding San Juan River

Space station astronauts photographed the stunning, horseshoe-shaped meanders of the San Juan River in Utah [1]. Millions of years ago, tectonic forces lifted the plateau, speeding up the water [1].



San Juan River meanders, photographed from the ISS on 6 October 2026 [1]

NASA astronaut photograph ISS075-E-188830

300 m
depth of the canyon carved by the river

9.81
Earth gravity acceleration in m/s^2 [4]

2,943 kJ
potential energy of 1 ton of water at 300 m

Rivers shape the face of our planet by moving water and sediment across landscapes [2]. When tectonic uplift raises a region, rivers gain steepness and erosive power [1]. The water cuts downward through 300-million-year-old rock, forming deep canyons and dramatic meanders known as goosenecks [1].

Water is vital for ecosystems and climate worldwide [3]. Understanding how flowing water stores and releases energy helps scientists protect river basins and manage water resources safely [2].

TIMELINE

Millions yrs
The San Juan River flows across a relatively flat plain.

Millions yrs
Tectonic forces uplift the Colorado Plateau, speeding up the river.

Present day
The river carves deep meanders into 300-million-year-old rock.

STOP 2 · THE PHYSICS

+20 XP FOR SOLVING

Potential energy of water

Water perched high on a plateau holds gravitational potential energy due to its height above sea level. When water flows downward, this stored energy converts into motion and erosion.

The energy depends on three factors: mass, gravity, and height. More mass or a higher drop gives the river greater potential energy to carve stone.

THE KEY EQUATION

$$Ep = m \times g \times h$$

Gravitational potential energy equation

Ep	potential energy	joule (J)
m	mass	kilogram (kg)
g	gravitational field	m/s^2
h	height / depth	meter (m)

Example masses, not measurements; use canyon depth as a drop; ignore friction. Gravity: 9.81 m/s^2 [4].

WORKED EXAMPLE

m = 1,000 kg, h = 300 m, g = 9.81

Ep = 1,000 × 9.81 × 300

Multiply: Ep = 2,943,000 J

Ep = 2,943 kJ

Water Mass	Depth h	Energy Ep
1 ton	300 m	2,943 kJ
2 tons	300 m	5,886 kJ
5 tons	300 m	14,715 kJ

Potential energy increases directly with the mass of water.

Potential energy for different water masses GRAPH



Heavier water masses store more gravitational energy at the same height.



BULB BUDDY SAYS

Higher drops and heavier water streams give rivers more energy to carve deep canyons over time.

STOP 3 · YOUR MISSION

+30 XP TO EARN

River energy mission

QUICK QUIZ · 3 QUESTIONS · TICK ONE BOX

Answer key: 1B · 2A · 3C

Q1 What determines gravitational potential energy in Ep = m × g × h? +5 XP

- A Temperature and wind
- B Mass, gravity, and height
- C Speed and time
- D Pressure and volume

Q2 In the equation Ep = m × g × h, what does h represent? +5 XP

- A Height or vertical drop
- B Mass of water
- C Gravity acceleration
- D Total energy

Q3 Which factor gives a river more erosive energy to carve canyons? +5 XP

- A Slower water flow
- B A flat dry desert plain
- C Steeper drop and heavier flow
- D Narrower riverbanks only

MINI LAB · TRY AT HOME OR SCHOOL

Water flow model

- [ ] Place a small tray or tilted board covered with sand or soil.
- [ ] Pour a cup of water steadily down the slope.
- [ ] Observe how the flowing water carves channels and meanders.

How does water flow carve channels in the sand? Relate it to energy.

BADGE UNLOCKED

River Explorer

You learned how water stores energy and carves canyons.

0 / 60 XP

Meander
A winding, horseshoe-shaped curve in a river channel.

Potential energy
Stored energy an object has due to its position or height.

Erosion
Wearing away of rock and soil by flowing water.

Sources
[1] NASA Earth Observatory, Taking a Gander at Utah's Goosenecks, 6 Nov 2026
[2] USGS, Rivers, Streams, and Creeks, 17 Sep 2026
[3] WMO, Water, 17 Jul 2026
[4] NASA Glenn Research Center, Weight and Mass, n.d.
Story: NASA ISS075-E-188830 (public domain). Cover: Illustration by ELIOS.